Digital Signal Processor

CS Department

Bonus Task (2 Marks) (Individual)

Delivery date: Practical Exam

Rules:

- Cheating is not allowed (across team, or across teams, any online resource).
- There will be discussion for this task at your practical exam.

Detection of periodic signal buried in noise. Given a sine wave signal (amplitude and frequency of your choice and 100 samples on x-axis).

- Apply auto-correlation of the sine wave signal and plot the output.
- Generate AWGN signal (additive white gaussian noise signal) of equal number of samples (100).
- Plot the noise signal.
- Add the sine wave signal (generated previously) to the noisy signal and plot the graph showing the corrupt signal.
- Now, apply auto-correlation of the sine + noise signal and plot the output.